A3 Problems: First algebra

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No calculator unless stated. For every equation, **check** your solution. For word problems, say what your letter stands for, write the equation, solve it, and answer the question in a sentence.

Practice

1. Simplify (*herleid*).

(a) $4a + 3b - 2a - 5b$
(b) $5x \cdot (-2x)$
(c) $3ab - 5ba + 2a$
(d) $2x^2 - 3x + 4x^2 + x - 7$
(e) $-3p \cdot 4q \cdot (-p)$
(f) $a \cdot a \cdot a + 2a^3$

2. Expand the brackets (*werk de haakjes weg*) and simplify.

(a) $4(3x - 2)$
(b) $-3(2 - 5a)$
(c) $7 - (4 - 3x)$
(d) $2(x + 5) - 3(x - 1)$
(e) $x(2x - 3) - x^2$
(f) $(x + 2)(x + 7)$
(g) $(2a - 1)(a + 4)$
(h) $(x - 3)^2$

3. Solve, and check your answer.

(a) $4x + 7 = 31$
(b) $7 - 2x = 3x - 8$
(c) $6x - 5 = 2x + 9$
(d) $4(x + 1) - 2(x - 3) = 20$
(e) $\dfrac{x}{4} + 2 = 5$
(f) $\dfrac{x + 3}{2} - \dfrac{x - 1}{3} = 3$

4. A plumber charges a €25 call-out fee plus €40 per hour.

(a) Write a formula for the cost K (in euros) of a job of t hours.
(b) Make a table for $t = 0, \frac{1}{2}, 1, 2, 3$ and draw the graph.
(c) What does a job of $2\frac{1}{2}$ hours cost?
(d) A bill is €165. How long did the job take?

5. The sequence $7, 11, 15, 19, \ldots$ continues in the same way.

(a) Give a direct formula for term number n (starting at $n = 1$).
(b) What is term 50?
(c) Is 403 a term of the sequence? Is 500?

6. Square tables are put in a row. One table seats 4 people, two tables in a row seat 6, three seat 8.

(a) How many people can sit at n tables in a row? Explain your formula with a picture.
(b) How many tables are needed for 30 people?

7. Solve the inequalities and show the solution on a number line.

(a) $5x + 2 > 17$
(b) $4 - 3x \leq 10$
(c) $2(x - 1) \geq 5x + 4$
(d) $\dfrac{x}{3} - 1 < 2$

8. Take $a = -3$ and $b = 2$. Calculate

(a) $a^2 - 2b$ (b) $-a^2 + ab$ (c) $(a + b)^2$ (d) $3a - 2(b - a)$ (e) $\dfrac{a - b}{a + b}$

School-exam level

9. Phone plan A costs €12 a month plus €0.10 per minute of calling. Plan B costs €20 a month plus €0.05 per minute.

(a) Write a formula for the monthly cost of each plan, with m the number of minutes.
(b) For how many minutes do the plans cost the same? Solve with an equation.
(c) Draw both graphs in one figure for $0 \leq m \leq 300$. Which plan is cheaper for someone who calls 250 minutes a month?

10. Anna is 4 years older than her brother. In 5 years their ages will add up to 40. How old are they now?

11. A rectangle has perimeter (*omtrek*) 46 cm. Its length is 5 cm more than its width. Find the length and the width.

12. The sum of four consecutive whole numbers is 154. Which numbers are they?

13. Two cyclists start at the same time from towns A and B, 60 km apart, and ride towards each other. The cyclist from A rides at 18 km/h, the one from B at 22 km/h. After how long do they meet, and how far from A?

14. Biologists noticed that crickets chirp faster when it is warm. A rule of thumb is $N = 7T - 30$, where N is the number of chirps per minute and T the temperature in °C.

(a) How many chirps per minute at 25 °C?
(b) Rewrite the formula as $T = \ldots$.
(c) You count 110 chirps per minute. What is the temperature?
(d) Below what temperature does the formula say the crickets stop chirping? Round to one decimal.

15. Use Gauss's trick (notes §6) to calculate:

(a) $1 + 2 + 3 + \cdots + 200$
(b) $2 + 4 + 6 + \cdots + 200$
(c) $3 + 7 + 11 + 15 + \cdots + 199$

16. A money box contains 25 coins, all 20-cent and 50-cent coins, worth €8.60 in total. How many of each are there?

17. A class trip costs €450 for the bus plus €12 entry per pupil. Each pupil pays €25. How many pupils must go for the money to cover the costs? Solve with an inequality.

18. A linear relationship has $y = 11$ when $x = 2$, and $y = 20$ when $x = 5$.

(a) Find the formula $y = ax + b$.
(b) Does the point $(10, 36)$ lie on the graph?
(c) For which x is $y = 0$?

19. (a) Solve $3(x - 2) + 6 = 3x$. What do you notice?

(b) Solve $4x - 2(x+1) = 2x + 5$. What do you notice?
(c) For which number c is every x a solution of $5x + c = 5(x+2)$? What happens for other values of c?

Challenge

20. (a) Calculate 1, $1+3$, $1+3+5$, $1+3+5+7$. Guess a formula for $1+3+5+\cdots+(2n-1)$.

(b) Prove your formula with Gauss's trick.
(c) Prove it again with a picture of dots arranged in a square.

21. Find all integers n (positive, negative or zero) for which $\dfrac{2n+7}{n+1}$ is an integer. (Hint: write $2n+7 = 2(n+1) + ...$)

22. (a) "Think of a number. Multiply it by 4. Add 12. Divide by 2. Subtract twice the number you started with." Prove that the answer is always 6.

(b) "Think of a number. Add 3. Multiply by the number you first thought of minus 3. Add 9." Prove that the result is always the square of your number.
(c) Invent a trick whose answer is always 5, and prove that it works.

23. (a) Explain with a picture of a square of side $a+b$ why $(a+b)^2 = a^2 + 2ab + b^2$.

(b) Use it to calculate 103^2 and 98^2 in your head.
(c) Use $(a+b)(a-b) = a^2 - b^2$ to calculate $97 \cdot 103$ in your head.
(d) A student writes $(a+b)^2 = a^2 + b^2$. Find a picture or a pair of numbers that shows him what he forgot.

24. Pens cost €3 and notebooks €5. You want to spend exactly €41. Find all possibilities (whole numbers, possibly zero, of each).

25. (a) Show that the sum of the first n terms of the arithmetic sequence $a, a+d, a+2d, ...$ is $\dfrac{n\,(2a + (n-1)d)}{2}$.

(b) Show that 2027 can be written as a sum of (at least two) consecutive positive whole numbers.
(c) Prove that 64 cannot be written as a sum of at least two consecutive positive whole numbers. (Hint: if the numbers are $m, m+1, \ldots, m+k-1$, then by (a) $k(2m+k-1) = 128$. Look at odd and even.)

26. Take any three-digit number with digits a, b, c (so the number is $100a + 10b + c$), and reverse it. Prove that the difference between the number and its reverse is always divisible by 99. For which numbers is the difference 0?

27. You drive 60 km to your grandmother at 30 km/h and drive back at 60 km/h.

(a) What is your average speed for the whole trip? (It isn't 45 km/h!)
(b) Show that for any distance d there and back, at speeds u and w, the average speed is $\dfrac{2uw}{u+w}$.
(c) How fast would you have to drive back to make the average 60 km/h?

28. (a) Two numbers have sum 20 and difference 6. Find them without trial and error.

(b) Two numbers have sum S and difference D. Find a formula for each number.

29. (a) Solve $3(x-2) - 2(x+1) > x - 10$. What happens?

(b) Give an inequality with no solutions at all, and explain why it has none.